

PROTOCOL A
Phase II Study of GLX-201 in Adults with Type 2 Diabetes Mellitus
Protocol Number: GLX-201-T2D-P2
Version: 1.0
Date: 09 March 2026
Sponsor: Global Metabolic Therapeutics Inc.

1. Study Synopsis	
Item	Description
Study Title	Phase II Randomized Double-Blind Study Evaluating the Safety and Efficacy of GLX-201 in Adults with Type 2 Diabetes Mellitus
Phase	Phase II
Study Population	Adults with Type 2 Diabetes Mellitus with suboptimal glycemic control
Sample Size	Approximately 120 participants
Duration	16 weeks treatment
Primary Endpoint	Change in HbA1c from baseline to Week 16

2. Background and Rationale
Type 2 Diabetes Mellitus (T2DM) is a chronic metabolic disorder characterized by insulin resistance and progressive pancreatic beta-cell dysfunction. Despite multiple therapeutic options, many patients fail to achieve sustained glycemic control. GLX-201 is an investigational small molecule designed to improve insulin sensitivity and reduce hepatic glucose output. Early Phase I studies demonstrated favorable pharmacokinetics and a promising safety profile. This Phase II trial aims to evaluate the efficacy and safety of GLX-201 in patients with T2DM with inadequate glycemic control despite stable background therapy.

3. Study Objectives
3.1 Primary Objective
To evaluate the change in glycated hemoglobin (HbA1c) from baseline to Week 16 in participants treated with GLX-201 compared to placebo.
3.2 Secondary Objectives

- Evaluate change in fasting plasma glucose
- Evaluate renal safety markers
- Evaluate incidence of treatment-emergent adverse events
- Assess body weight changes

4. Study Design
This is a **multicenter, randomized, double-blind, placebo-controlled Phase II trial**. Participants meeting eligibility criteria will be randomized **1:1** to receive:

- GLX-201
- Matching placebo

Both groups will continue standard background therapy.

5. Study Population

Adults with **Type 2 Diabetes Mellitus** with HbA1c above recommended target levels despite stable therapy.

6. Inclusion Criteria

Participants must meet **all** the following criteria:

1. Age **18 to 75 years**
2. Confirmed diagnosis of **Type 2 Diabetes Mellitus**
3. HbA1c between **7.0% and 10.0%**
4. Estimated glomerular filtration rate (**eGFR ≥ 45 mL/min/1.73m²**)
5. Stable anti-diabetic medication regimen for **≥ 12 weeks**
6. Body mass index (**BMI 25–40 kg/m²**)
7. Ability to understand and sign **informed consent**
8. Willingness to comply with study procedures and visits

7. Exclusion Criteria

Participants will be excluded if they meet any of the following:

1. Diagnosis of **Type 1 Diabetes Mellitus**
2. **eGFR < 45 mL/min/1.73m²**
3. History of **major cardiovascular event within 3 months**
4. Active **liver disease** or hepatic dysfunction
5. Participation in **another interventional clinical trial within 30 days**

8. Study Procedures and Timeline

Visit	Timepoint	Procedures
Screening	Day -28 to Day -1	Eligibility, labs, consent
Baseline	Day 0	Randomization, labs
Visit 2	Week 4	Safety labs, vitals
Visit 3	Week 8	HbA1c, labs
Visit 4	Week 12	Safety evaluation
Visit 5	Week 16	Final efficacy assessment
Follow-Up	Week 18	Safety follow-up

9. Endpoints

Primary Endpoint

Change in **HbA1c from baseline to Week 16**

Secondary Endpoints

- Change in fasting glucose
- Renal function markers
- Change in body weight
- Treatment emergent adverse events